

**Guangdong Technion**

Israel Institute of Technology

广东以色列理工学院

Homework 7 - Deadline December 20th

Exercise 1. a) 10 points. Solve $\int \frac{1}{(x^2 + 4)^2} dx$.

1. b) 15 points. Compute

$$\int \frac{x^4 + 8x^2 + x + 16}{x(x^2 + 4)^2} dx.$$

Exercise 2. 25 points. The region $\mathcal{R}$ enclosed by the curves $x = e^y$, $y = 0$, $x = e$ and $x = e^3$ is rotated about the **x-axis**. Find the volume of the resulting solid.

Exercise 3. 25 points. The region $\mathcal{R}$ enclosed by the curves $y = \cos(x^2)$, $x = 0$, $y = 0$ and $y = \frac{\sqrt{2}}{2}$ is rotated about the **y-axis**. Find the volume of the resulting solid.

Exercise 4. a) 10 points. Solve

$$\int \frac{(2 \tan^2(x) + 1) \sec^2(x)}{(\sec^2(x) - 1)(\tan^2(x) + 1)} dx$$

b) 15 points Use 1st a substitution and then integration by parts, to solve

$$\int \frac{\sin\left(\ln \sqrt{1 + x^{-4/5}}\right)}{(1 + x^{-4/5})(x + x^{9/5})} dx$$